NLP Project: Topic Modeling of the 1981 French Legislative Elections

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Abstract

This report details the implementation of NLP techniques to calculate theme distributions across political parties using the Archelec corpus from the 1981 French legislative elections. We describe the data retrieval process, the statistical properties of the corpus, and leverage the BERTopic framework to visualize the intersection of political discourse and party affiliation. Our findings highlight the thematic distinctness of major political entities during this period.

1 Problem Definition

Topic modeling is a technique in NLP that helps to identify themes within large texts. It is primarily used to automatically identify topics in a collection of documents without the need for manual annotation.

- **The Goal:** Automatically extract the main political themes of the 1981 election and quantify how these topics are distributed across specific political parties.
- **Dataset:** The *Archelec* archive, specifically focusing on the 1981 Legislative election profession de foi (manifestos).

2 Data Retrieval

2.1 Preprocessing

The raw text underwent standard NLP cleaning: removal of punctuation, lowercasing, and the elimination of French stop words.

2.2 Statistical Description

The following metrics describe the processed corpus:

- **Total Documents:** 3,182 manifestos.
- **Average Characters Count:** 4138 characters on average per text before cleaning, 2573 characters on average after cleaning. 801 tokens before cleaning, 303 after.
- **Class Balance (Parties):** The distribution is highly skewed, dominated by major parties like the PS, RPR, and UDF, while dozens of fringe parties contribute single-document entries.

3 State of the Art

Topic modeling has evolved from purely statistical counts to deep semantic understanding:

- **Classical:** Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF). These rely on "bag-of-words" and often struggle with short texts or nuanced context.
- **Modern:** Neural models such as Top2Vec and BERTopic, which utilize transformer-based embeddings to capture semantic meaning rather than just word frequency.

4 Proposed Model

4.1 Model Selection

For this task, we selected **BERTopic**, a transformer-based neural topic modeling framework.

4.2 Technical Workflow

BERTopic extracts latent themes from the text:

1. **Embeddings:** Documents are converted into dense, high-dimensional semantic vectors. We utilized `dangvantuan/sentence-camembert-base`, a model specifically optimized for the French language, to capture the historical and political nuances of the 1981 electoral texts.
2. **Dimensionality Reduction:** Because clustering in high dimensions is computationally expensive and prone to the "curse of dimensionality," **UMAP** (Uniform Manifold Approximation and Projection) is applied. This reduces the vectors to a lower-dimensional space while preserving their local and global geometric structures.
3. **Clustering: HDBSCAN** (Hierarchical Density-Based Spatial Clustering of Applications with Noise) groups the reduced vectors into dense clusters (topics).
4. **Topic Representation:** Finally, BERTopic extracts the most representative words for each cluster using a class-based variation of TF-IDF.

4.3 Justification

The choice of BERTopic over classical models like Latent Dirichlet Allocation (LDA) or Non-negative Matrix Factorization (NMF) was driven by three specific requirements of our corpus and analytical goals:

- **Semantic Superiority:** Traditional models rely strictly on word frequencies ("bag-of-words") and struggle with synonyms or varied phrasing. By using contextual embeddings (CamemBERT), BERTopic understands the actual meaning of the sentences. It can recognize that two manifestos discussing an economic policy belong to the same theme, even if they use entirely different vocabulary.
- **Class-Based Analysis for Political Parties:** The primary goal of this project is to analyze theme distributions *across different parties*. BERTopic's c-TF-IDF is specifically designed for this. It allows us to seamlessly group documents by their political party (class) and extract the unique vocabulary that distinguishes one party's discourse from another.

5 Experiments

Detail the environment and the specific configuration used for the topic modeling pipeline:

- **Infrastructure & Libraries:** The project was implemented in Python. Data manipulation and text extraction were handled using `pandas` and `pathlib`. The core NLP pipeline relied on `Sentence-Transformers` for embeddings, `scikit-learn` (`CountVectorizer`) for text vectorization, and the `BERTopic` library. Visualizations were produced using both Plotly (for top words) and `matplotlib` (for thematic distribution).
- **Embedding Model:** To capture the specific semantic and syntactic context of the French language, documents were embedded using `dangvantuan/sentence-camembert-base`. This CamemBERT-based model is highly optimized for French sentence-level representations.
- **Hyperparameters & Configuration:**
  - *Vectorization:* A list of French stop words (`stop_word_fr.txt`) was passed to the `CountVectorizer` to clean the corpus of non-semantic noise prior to clustering.

- *BERTopic settings:* The language parameter was set to `"multilingual"`, and document-topic probability calculations (`calculate_probabilities`) were set to `False` to optimize memory consumption and execution speed.

- **Filtering Parameters:** To ensure robust statistical representation and readable visualizations, the final analysis was restricted to the 10 most frequently occurring topics (Topics 0 to 9) and the top 10 largest political parties.

6 Analysis and Conclusion

6.1 Results

Below are the visualizations representing the topic distributions and the thematic clusters identified by the model.

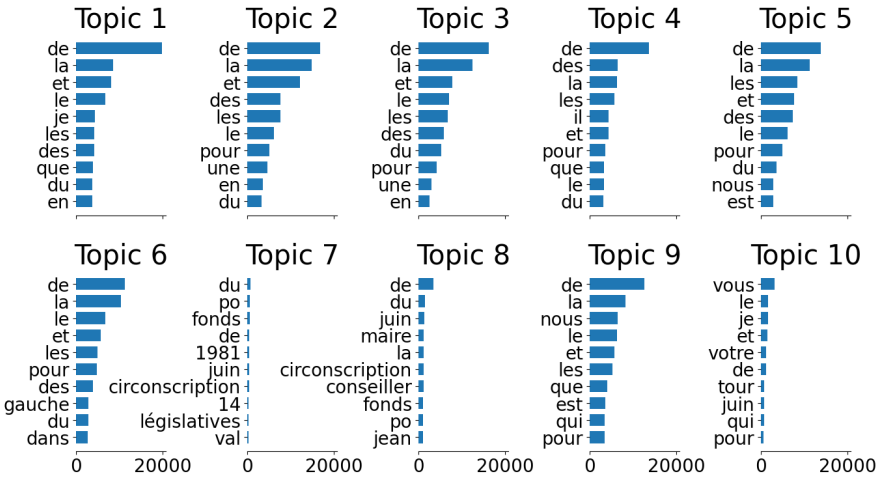

Figure 1: Top words (LDA) without preprocessing

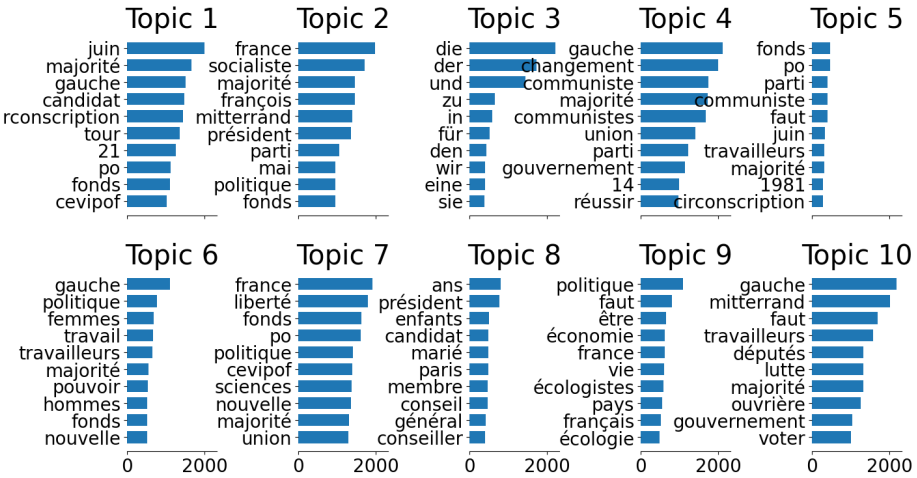

Figure 2: Top words (LDA) with stop words

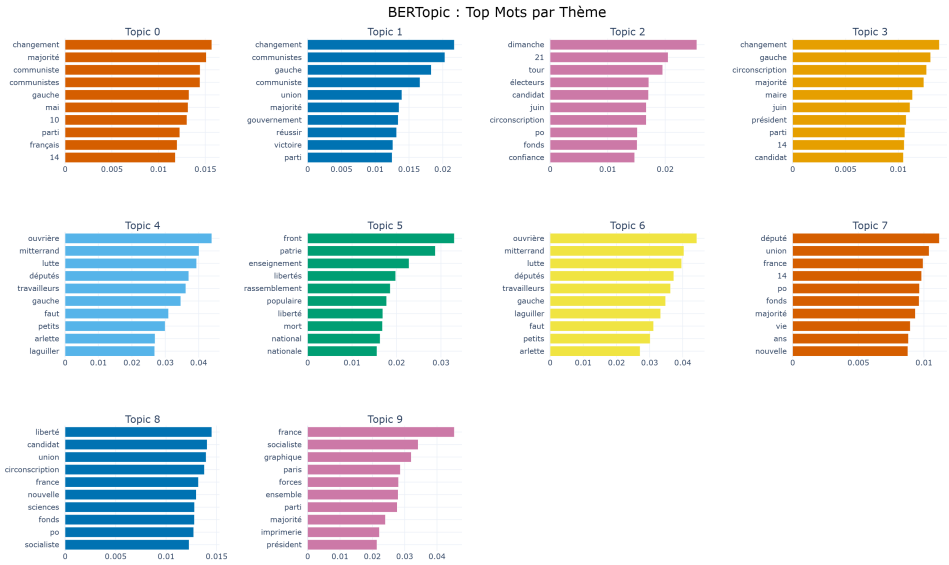

Figure 3: 10 Topics (Bertopic)

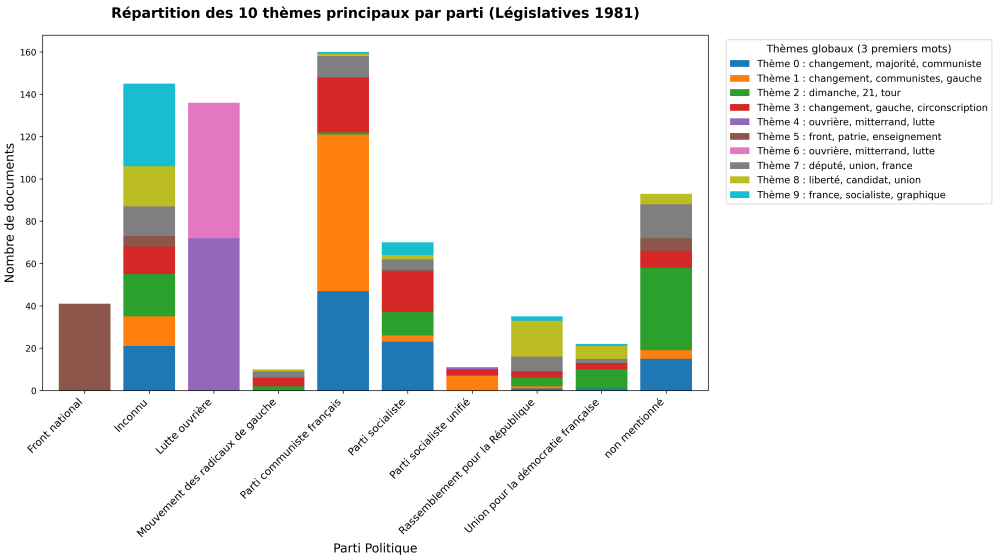

Figure 4: Distribution of Topics per Party: 10 most prominent political groups.

6.2 Analysis

The model successfully identified core 1981 themes. The themes are clearly different between parties, and they match what we would expect from them. For exemple, Lutte ouvriere focuses entirely on topics 4 and 6 (lutte, Mitterand, ouvriere), while Front National focuses on theme 5 (front, patrie, enseignement). Other parties like PS are mixing a lot of different themes. RPR and UDF have the same themes associated to them. It is important to note that often the party is not mentionned (or not known), so themes are diverse. This can be interesting to see to what parties those who do not have a party are the closest to.

6.3 Conclusion

BERTopic proved highly effective for this corpus. Despite the age of the texts, transformer-based embeddings captured the ideological nuances of the 1981 campaign more effectively than traditional frequency-based models.

References

[1] Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv preprint arXiv:2203.05794.